



Not All Tokens Are Equally Useful for Steering: Robust Directions and Prefix Steering

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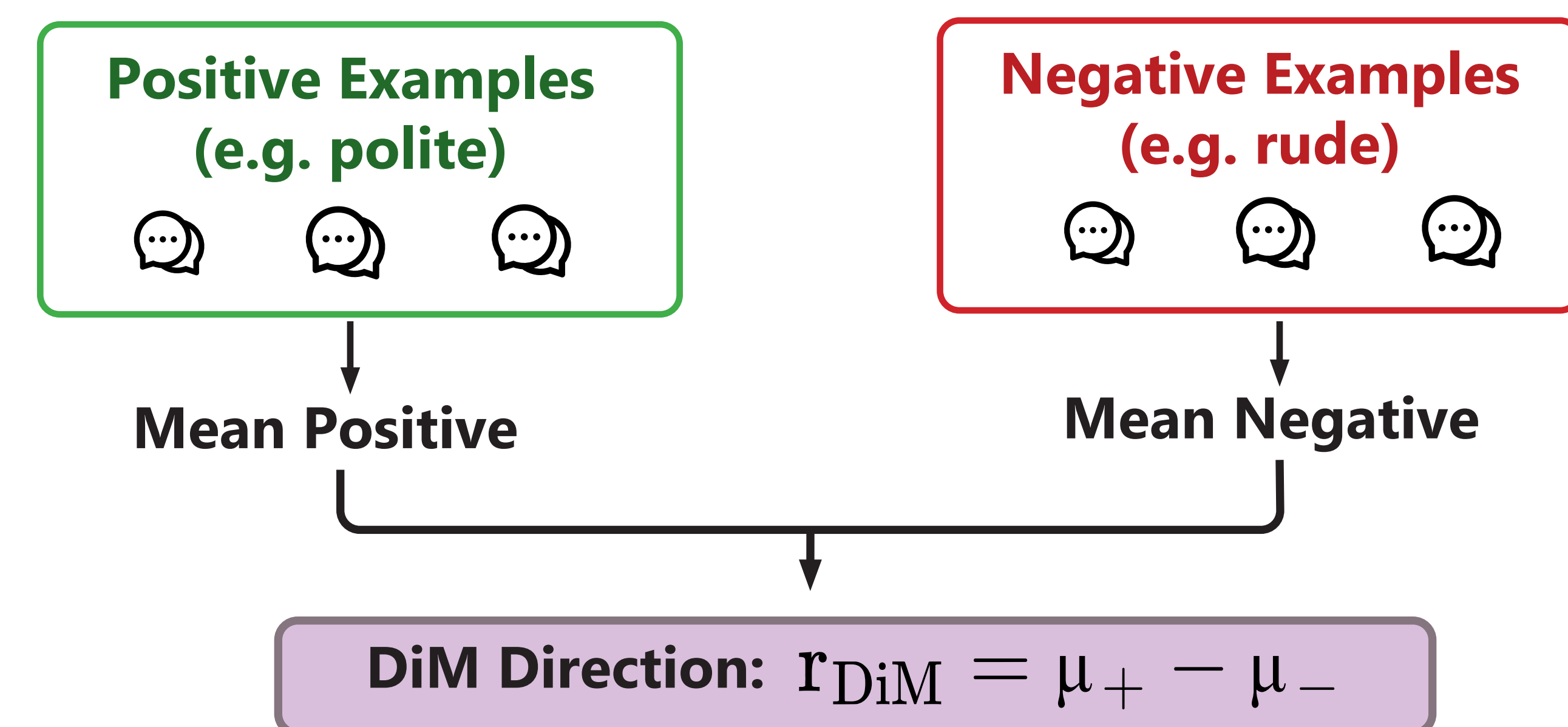
[Code](#)

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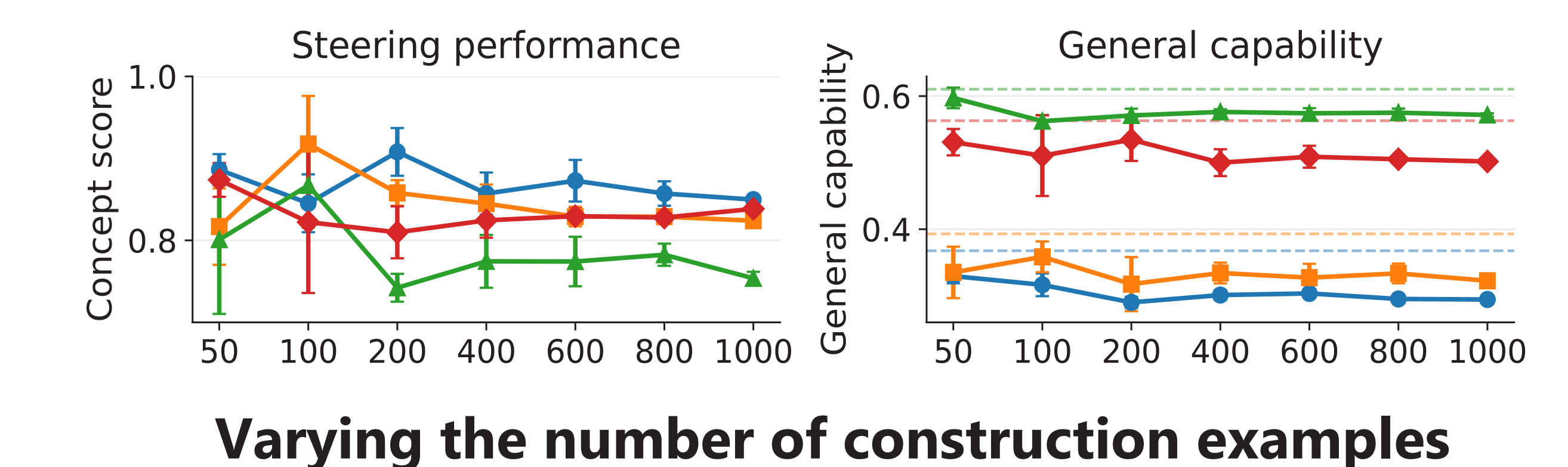
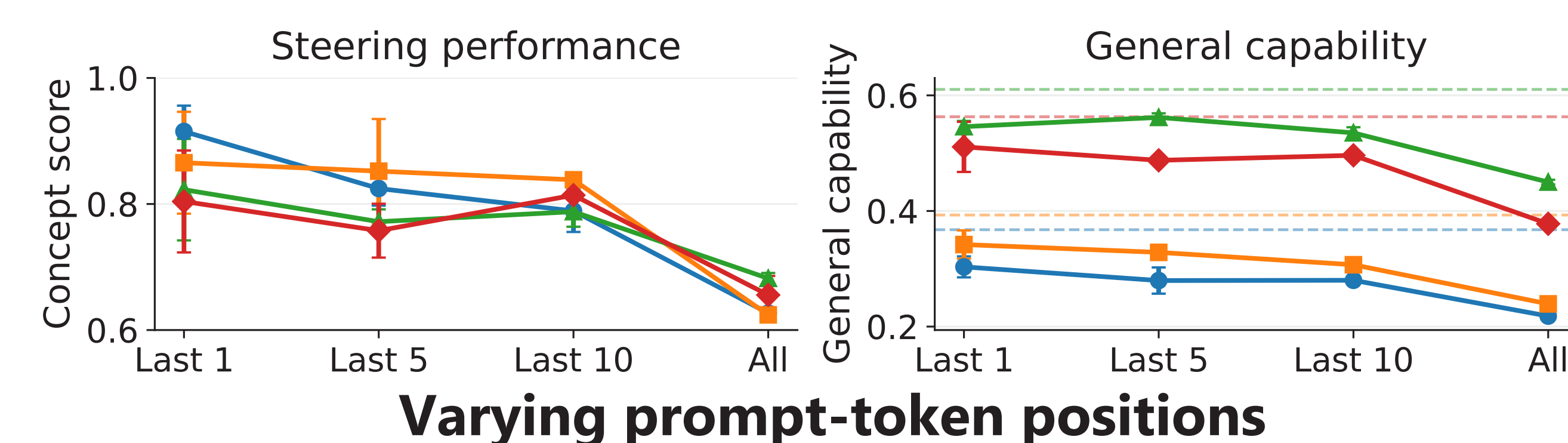
1 Motivation & Contribution

Activation steering controls LLM behavior by adding steering vectors to hidden states. However, standard methods such as **Difference-in-Means (DiM)** can be unstable and hurts general capabilities.



Key Questions

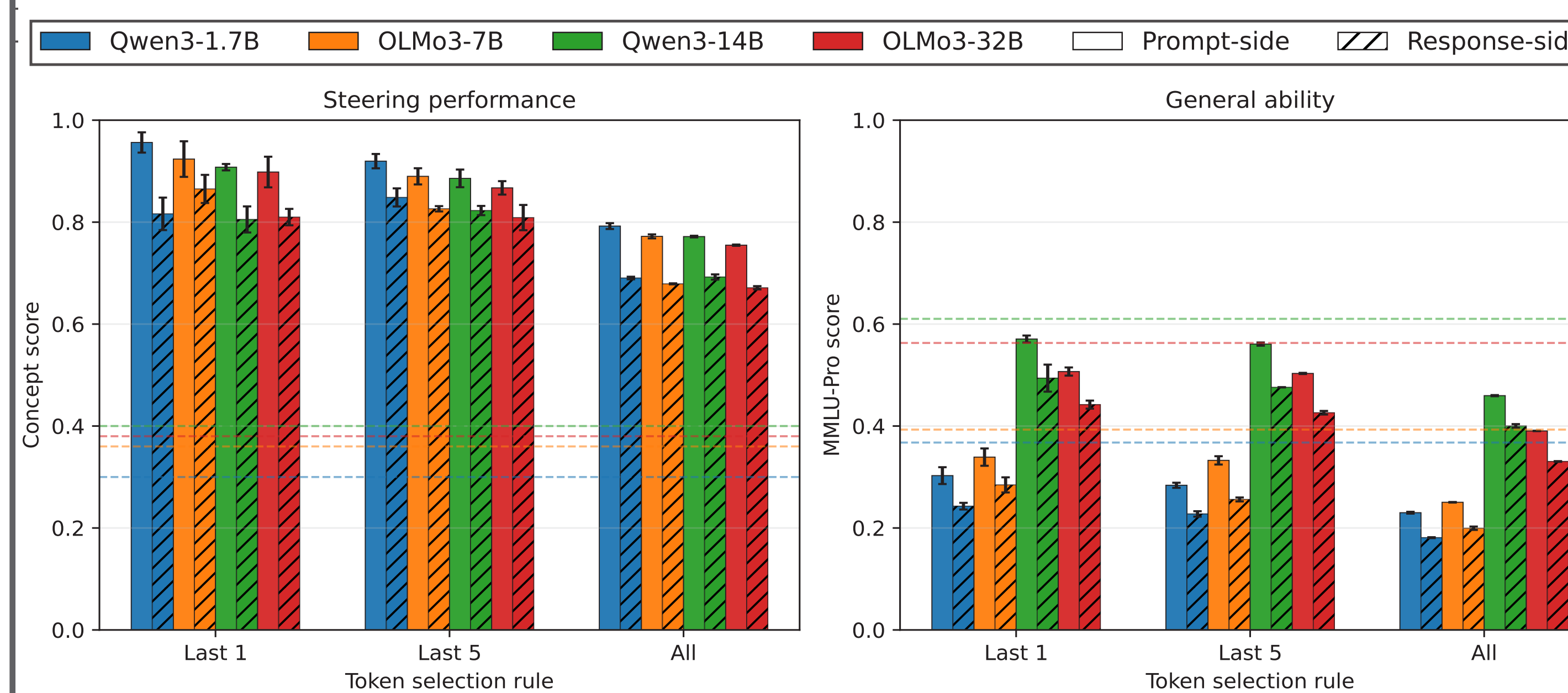
- Q1:** Which activations should be used to construct the direction?
- Q2:** which generated tokens should receive the steering intervention?



Takeaway: More is not always better; selective is better. Using more tokens does not necessarily improve steering performance, while selecting intervention-relevant positions can lead to more effective control.

2 Construction Matter: Robust DiM

Diagnosing Construction Choices



Insights

- Recent tokens outperform all tokens.
- More tokens improve directional stability, but not downstream performance.
- Prompt activations are more intervention-relevant

Robust DiM: Representative + Discriminative

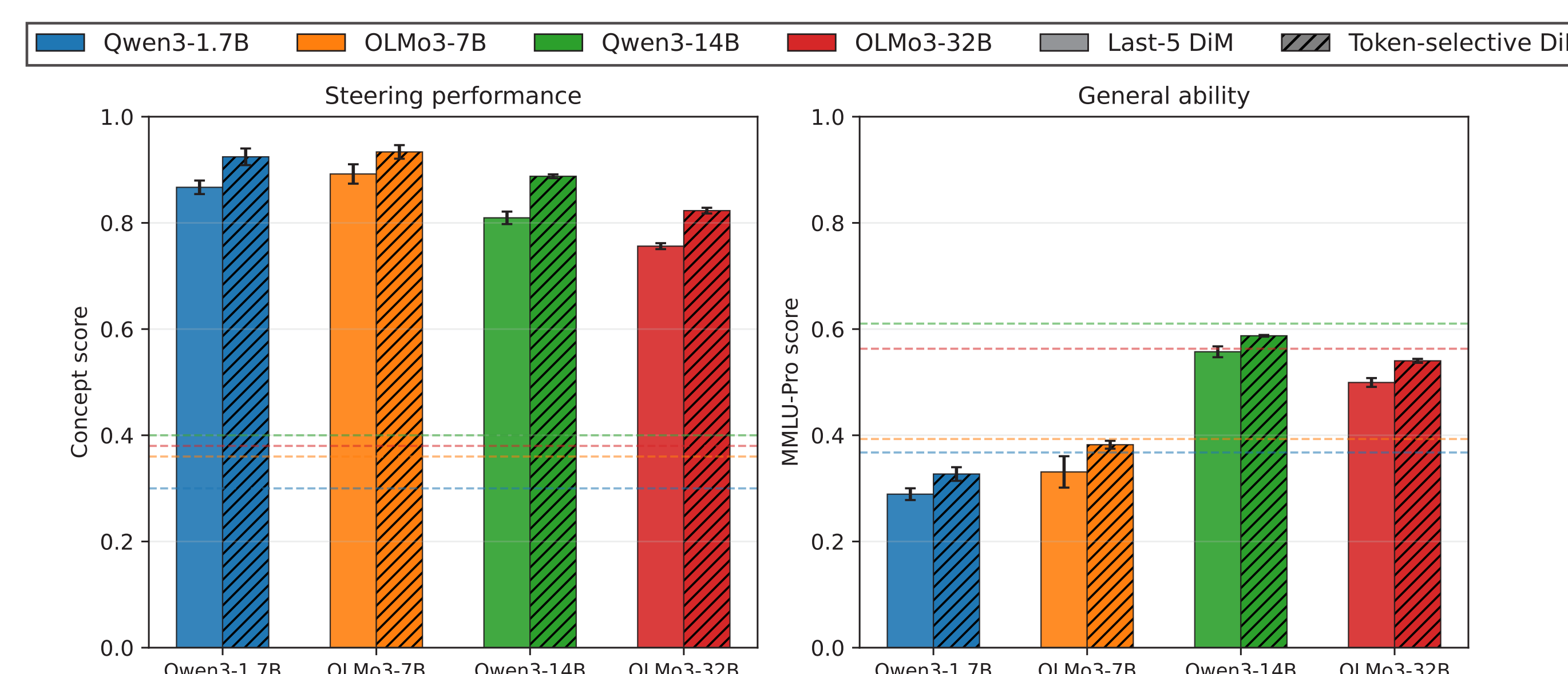
- Score each activation in the positive class:

$$s^+\left(\mathbf{h}\right)=\frac{\left\|\mathbf{h}-\mu^{+}\right\|_2^2-\left\|\mathbf{h}-\mu^{-}\right\|_2^2}{\left\|\mathbf{h}-\mu^{+}\right\|_2^2+\left\|\mathbf{h}-\mu^{-}\right\|_2^2}$$
- Negative class use similar ways.
- Select the top-K activations from each class to construct a robust steering direction.

Representative: close to its own class center.

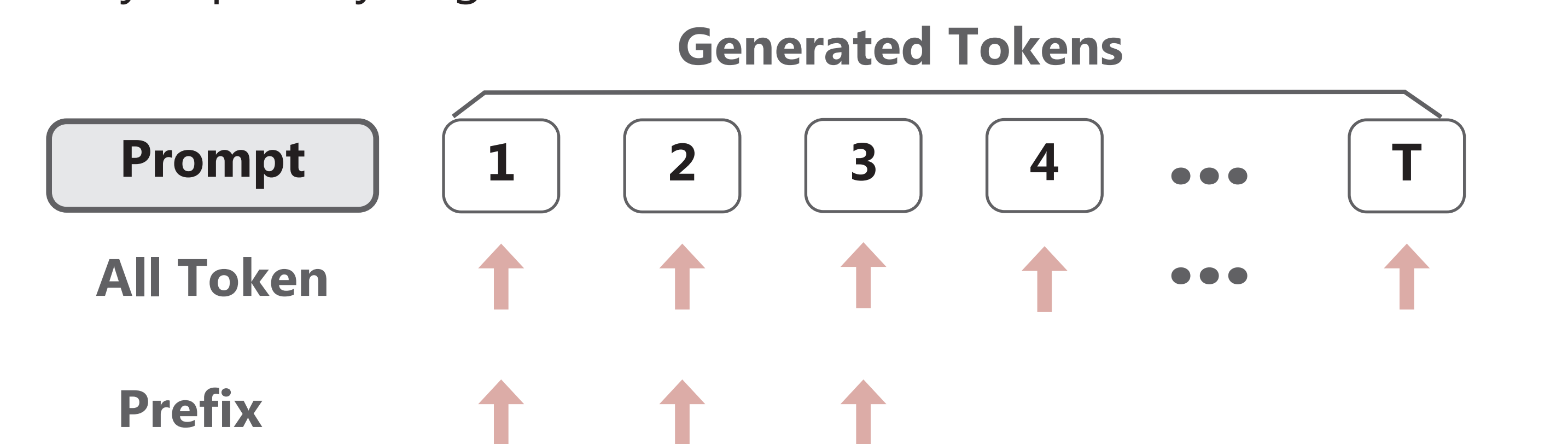
Discriminative: far from the opposite class center.

Robust DiM vs Standard DiM

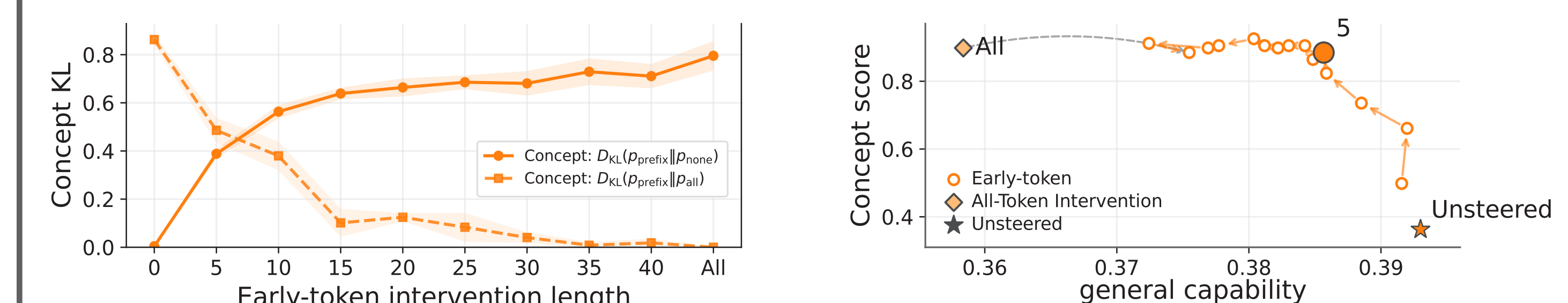


3 One Token Is Enough: Prefix Steering

Prefix steering only intervenes on the **first few tokens**, and often **sufficient** to recover most effect while reducing unnecessary capability degradation.



Effect of Prefix Length (Example: OLMO3-7B)



Joint Effect of steering strength and prefix length (Example: OLMO3-32B)



Key Finding

- Most steering effect emerges within the first ~5 tokens
- Later interventions mainly cause distributional shift and hurt general ability.
- Longer steering improve target control but hurts general ability.
- Short but stronger prefixes (most time **1-token is enough**) recover most steering gains with less general degradation.

Effective activation depends on selecting both intervention-relevant construction activations and intervention-relevant generation position, rather than uniformly averaging or steering all tokens.